How FlowState makes a forecast

IBM · 18.5M parameters (r1.1) · state-space encoder, functional decoder · zero-shot

1. Every step a token

FlowState reads one series. There are no patches: each step becomes its own token.

past *future*

Target temperature

Each value is scaled with the mean and standard deviation of the steps up to it, then a linear layer turns value and missing flag into 512 numbers.

2. A clock for any rate

The scale factor s tells the model how many steps make one season, so the same weights fit daily or monthly data.

$$s = \frac{24}{m}$$

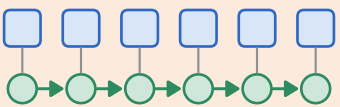
m is the number of steps in one season: 365 for days, 12 for months, 252 for trading days.

3. Six state-space layers

No attention. Each S5 layer updates a hidden state at every step; its step size Δ is multiplied by s.

$$h_t = \bar{A}\, h_{t-1} + \bar{B}\, x_t$$

one token per step



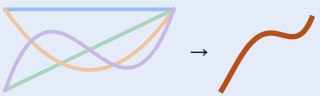
state updated every step

S5 layer: state of 512, then a sigmoid output gate

Feed forward: gated, 512 → 1,024 → 512, then LayerNorm; shortcut around the layer

4. A curve, not fixed steps

The decoder gives the weights of **256 Legendre curves** for each of **9 quantiles**. The sum is read off at each future step.



weighted sum of curves

median
60% band
80% band

The point forecast is a weighted mean of the 9 quantiles; the middle ones count most.

A continuous-time encoder and decoder let one set of weights serve any sampling rate.

6 S5 layers · width 512 · state 512 · 256 Legendre curves per quantile · trained on 4,096-step contexts

In the notebooks: last 2,048 days or full monthly history · s from the yearly season · each metal on its own · no covariates